Unit: 6 Unit 9: Data Statistical Analysis **Name:** ____________
Topic: Box Whisker (Dispersion) **Date:** ________

Challenge (Habanero)

Q1. A hiking group recorded the following distances: 15, 18, 22, 25, 30. What is the median?
(A) 20
(B) 22
(C) 25
(D) 28
(E) 30

Q2. A set of fuel consumption data has a lower quartile of 8 and an upper quartile of 12. What is the interquartile range?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10

Q3. A box-and-whisker plot shows a minimum value of 5, a maximum value of 25, and a median of 15. If the lower quartile is 10, what is the interquartile range?
(A) 5
(B) 10
(C) 15
(D) 20
(E) 25

Q4. The following data represents the number of birds spotted during a nature walk: 2, 4, 6, 8, 10, 20. Which value is an outlier?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 20

Q5. A set of data has the following five-number summary: $min = 5$, $Q_1 = 10$, $median = 15$, $Q_3 = 20$, $max = 30$. What percentage of the data is between 10 and 20?
(A) 25
(B) 50
(C) 75
(D) 90
(E) 100

Q6. The box-and-whisker plot below represents the fuel efficiency of a car in miles per gallon. If the lower quartile is 20 and the upper quartile is 30, what can be inferred about the fuel efficiency?
(A) Most of the data points are below 20
(B) Most of the data points are between 20 and 30
(C) Most of the data points are above 30
(D) The data is symmetric
(E) The data is skewed to the right

Q7. A hiker recorded the following times to complete a trail: 2, 3, 4, 5, 6, 7, 8, 9. If the times are represented in a box-and-whisker plot, which of the following is true?
(A) The median is 4.5
(B) The lower quartile is 3
(C) The upper quartile is 6
(D) The interquartile range is 3
(E) All of the above

Q8. A nature reserve has the following number of animals: 10, 12, 15, 18, 20, 25. What is the Q_3?
(A) 15
(B) 17
(C) 18
(D) 20
(E) 22

Open Ended Questions (FRQ)

Q9. A group of friends went on a road trip and recorded their fuel consumption. The data is as follows: 15, 18, 20, 22, 25, 30, 35. Create a box-and-whisker plot and identify any outliers.

Q10. The number of hikers on a trail for the past week is as follows: 20, 25, 30, 35, 40, 45, 50. Calculate the median, lower quartile, and upper quartile. Then, create a box-and-whisker plot and comment on the dispersion.

Chef's Tips

- Tip 1: Ensure students understand the concept of quartiles and interquartile range
- Tip 2: Provide examples of real-world applications of box-and-whisker plots
- Tip 3: Emphasize the importance of identifying outliers in a dataset